



Sequence & Series

A **sequence** is a set or collection of numbers arranged in an orderly manner such that the preceding and the following numbers are completely specified. An **infinite sequence** is a function whose domain is the set of positive integer. If the domain of the function consists of the first n positive integers only, then it is said to be a **finite sequence**. The numbers in the given sequence are called **elements**. An element is sometimes called **term**.

1, 3, 5, 7, 9 → Finite sequence
1, 3, 5, 7, 9, ... → Infinite sequence

Series is the sum of the terms in a sequence. An **alternating series** has positive and negative terms arranged alternately. If an infinite series has a finite sum, it is referred to as **convergent series** and **divergent series** if it has no sum at all.

PROGRESSION

A **progression** is simply another term for a sequence. The most common types of progression are:

1. Arithmetic Progression (AP)
2. Geometric Progression (GP)
3. Harmonic Progression (HP)

A sequence is said to be in **arithmetic progression** if its succeeding terms have a common difference. The sum of all the terms in arithmetic progression is called as **arithmetic series**.

Last term (nth term): $a_n = a_1 + (n-1)d$

Sum of all terms:

$$S = \frac{n}{2}(a_1 + a_n)$$

$$S = \frac{n}{2}[2a_1 + (n-1)d]$$

where:

a_1 = first term

a_n = last term (n^{th} term)

n = number of terms

d = common difference $= a_2 - a_1 = a_3 - a_2 = \dots$

A sequence is said to be a **geometric progression** if its succeeding terms have a common ratio. The sum of all the terms in geometric progression is called as **geometric series**.

Last term (nth term): $a_n = a_1 r^{n-1}$

Sum of all terms:

$$S = \frac{a_1(r^n - 1)}{r - 1}$$

$$S = \frac{a_1(1 - r^n)}{1 - r}$$

where:

a_1 = first term

a_n = last term (nth term)

n = number of terms

r = common ratio $= a_2/a_1 = a_3/a_2$

PROGRESSION continuation

Infinite geometric progression is geometric progression only that the number of terms (n) is extremely large or infinity.

If $r > 1$, sum of all terms is infinite

If $r < 1$, the sum of all terms is $S = \frac{a_1}{1-r}$

where:

a_1 = first term

r = common ratio

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A sequence of numbers whose reciprocals form an arithmetic progression is known as **harmonic progression**. In solving a problem, it would be wise to convert all given terms into arithmetic sequence by getting its reciprocals. Use the formulas in arithmetic sequence and take the reciprocal of resulting value to obtain the equivalent harmonic term for an answer.

Fibonacci and Related Sequences

The following are the some of the famous and related sequences:

1. **Fibonacci Numbers** – Named after the Italian merchant and mathematician, Leonardo di Pisa or Fibonacci (Figlio dei Bonacci, "Son of the Bonnaccis").

1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144...

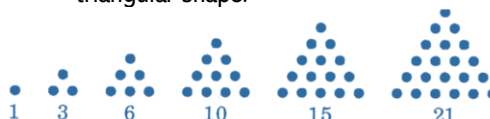
Each number is equal to the sum of the two preceding numbers.

2. **Lucas Sequence** – Named after Edouard Lucas (1841 – 1891). Like the Fibonacci numbers, every term of the Lucas sequence is the sum of the two preceding numbers.

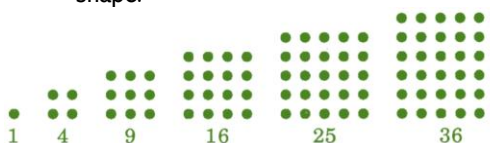
1, 3, 4, 7, 11, 18, 29, 47, 76, 123...

3. **Figurate Numbers:**

- I. **Triangular numbers:** Numbers which can be drawn as dots and arranged in triangular shape.



- J. **Square numbers:** Numbers which can be drawn as dots and arranged in square shape.

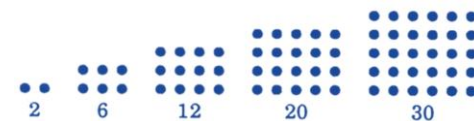


Fibonacci and Related Sequences continuation

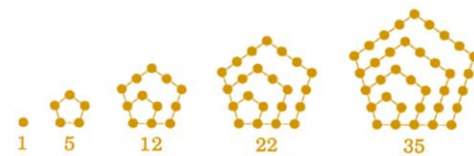
- C. **Gnomons:** Numbers which can be drawn as dots on equally long legs of a right angle.



- D. **Oblong numbers:** Numbers which can be drawn as dots and arranged in a rectangle shape.



- E. **Pentagonal numbers**



- F. **Cubic numbers**



- G. **Tetrahedral numbers**



- H. **Square pyramidal numbers**

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By piling up tetrahedral numbers, 1, 4, 10, 20, 35, 56, etc, we make four-dimensional numbers:

1, 5, 15, 35, 70,...